

EMM 30'2
By Dr. Bruno Mose.
QUESTION ONE (30 Marks) - COMPULSORY

(a) Differentiate between statically determinate and statically indeterminate pin jointed frames [2 marks]

(b) Using the method of resolution of forces at the joints, determine the axial forces of the members in a plane pin-jointed frames shown in Fig. Q. 1 (b). [7 marks]

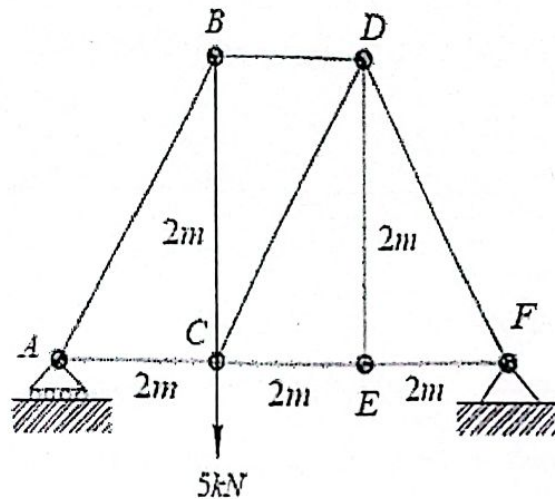


Fig. Q. 1 (b).

(c) A tension specimen of circular cross section tapers uniformly from 20mm to 16mm diameter over a gauge length of 200mm as shown in Fig. Q.1d. When an axial force of 40kN is applied, the extension measured on this gauge length is 0.4mm.

(i) Find the modulus of elasticity of the material. [3 marks]

(ii) Compare the strain energy in this specimen with that in a specimen of the same material of uniform diameter 18mm carrying the same load. [2 marks]

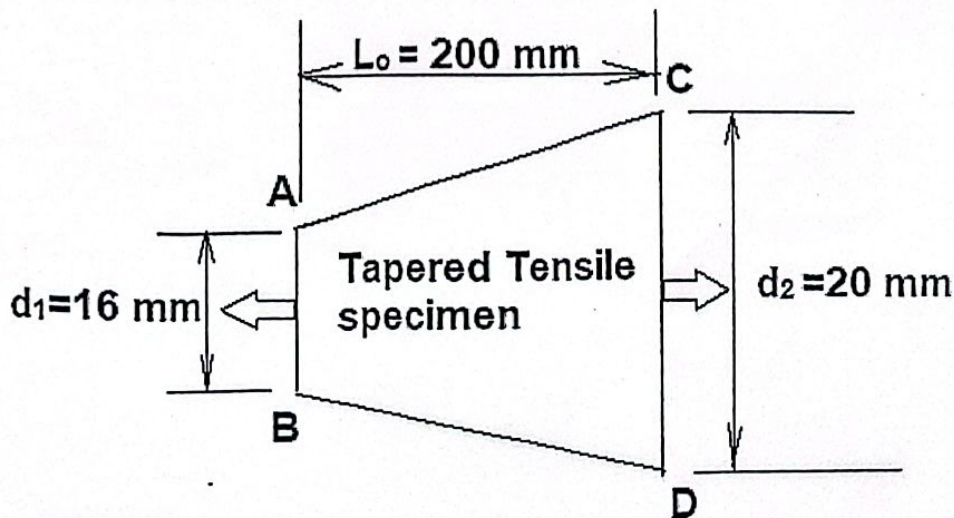


Fig. Q. 1d.

- (d) Show that the strain energy stored in a shaft of circular cross section transmitting torque is given as; [4 marks]

$$U = \frac{\tau_{max}^2 V}{4G}$$

where; U-strain energy G-Modulus of Rigidity τ_{max} -Maximum shear stress of the shaft material

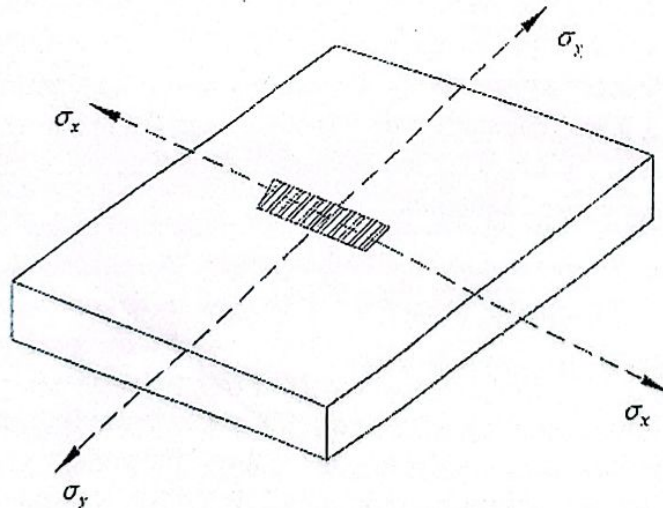
- (e) Show that, in a biaxial stress system, if ϵ_x and ϵ_y are the strains in two perpendicular directions, then the normal stresses in these directions are given by,

$$\sigma_x = \frac{E}{1 - \nu^2} [\epsilon_x + \nu \epsilon_y]$$

$$\sigma_y = \frac{E}{1 - \nu^2} [\epsilon_y + \nu \epsilon_x]$$

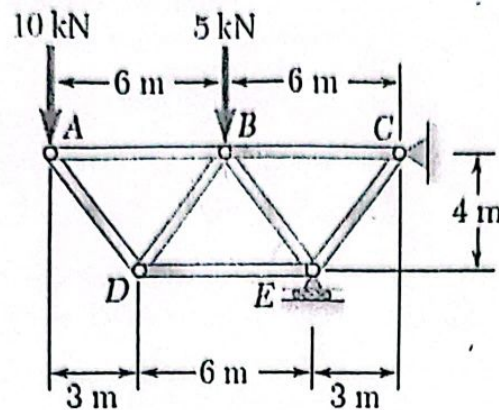
Where E and ν are the modulus of elasticity and Poisson's ratio of the material respectively. [4 marks]

- (f) A rectangular plate of uniform thickness has a strain gauge rosette bonded to one surface at the center as shown in the Fig. Q 1g. It is placed in a test rig which can apply a biaxial force system along the plate edges. If the measured strains are +0.0005 and 0.0007 in the x and y directions, ($E = 208 \text{ GN/m}^2$ and $\nu = 0.3$), find;
- The corresponding stresses set up in the plate. [2 Marks]
 - The strain through the plate thickness. [2 Marks]



Q 1g

- (g) Using the method of joints, determine the force in each member of the truss in Fig Q.1h
[4 marks]



Q 1h

QUESTION TWO (20 Marks)

- (a) Show that the Longitudinal stress in a thin-walled cylinder is given by;

$$\sigma_l = \frac{pr}{2t}$$

Where; r = Internal radius of pressure vessel, t = Wall thickness, p = fluid pressure inside the vessel.
[2 Marks]

- (b) Nairobi Coca Cola Bottlers wishes to test if their newly designed Novida packaging can is safe. They load the can with a 0.8MPa pressurized fluid and then bond two strain gauges, one in the longitudinal and the other on the circumferential directions. Once the can is popped open, these strain gauges record strains of 0.000102 and 0.0005 in the longitudinal and hoop directions respectively. The can is made from aluminium material whose Young's modulus E and Poisson's ratio ν is 60GPa and 0.33 respectively. Given that the internal diameter of the can is 80mm, calculate the wall thickness of the can. You may assume the following general equation;
[5 Marks]

$$\epsilon_x = \frac{1}{E} [\sigma_x - \nu(\sigma_y + \sigma_z)]$$

where all parameters have their usual meaning.

- (c) A group of fifth year students pursuing a Bachelor of Science degree in Petroleum Engineering at KU have developed a new aluminium alloy. One of the tasks ahead was to determine the four elastic constants (that is ν , E , K and G) for the material. They were advised by an Engineer from Kenya Bureau of Standards to subject a thin-walled cylinder made from the material to an internal pressure, and then measure the longitudinal and circumferential strains. They fabricated a 200mm diameter and 3mm wall thickness

cylinder from their material and subjected it to an internal pressure of 6MPa. Determine therefore, the four elastic constants of this new material if the strain gauges recorded strains of 0.00243 and 0.00057 at full pressure in the circumferential and longitudinal directions respectively. [8 marks]

The elastic constants are related as follows;

$$E = 2G(1 + \nu)$$

$$K = \frac{E}{3(1 - 2\nu)}$$

$$E = \frac{9GK}{3K + G}$$

where all parameters have their usual meaning.

- (d) Show that the volumetric strain of a bar of square cross sectional area undergoing axial loading is given by,

$$\epsilon_v = \epsilon_x(1 - 2\nu)$$

where ϵ_x is the longitudinal strain and ν is the Poisson's ratio.

[5 Marks]

QUESTION THREE (20 Marks)

- (a) A hollow steel shaft with an external diameter of 150 mm is required to transmit 1 MW at 300 rev/min. Calculate a suitable internal diameter for the shaft if its shear stress is not to exceed 70 MPa. Compare the torque-carrying capacity of this shaft with a solid steel shaft having the same weight per unit length and limiting shear stress. Take the value of $G = 80$ GPa. [8Mks]
- (b) A steel shaft ABCD having a total length of 1.5 m is made of three lengths AB, BC and CD each 0.5 m long as shown in Fig Q3b. AB and BC are solid, having diameters of 60 mm and 70 mm respectively, and CD is hollow having outside and inside diameters of 70 mm and 45mm respectively. When an axial torque of 2 kN.m is transmitted from one end of the shaft to the other, the total angle of twist from A to D is 1.20. Determine,
- (i) the maximum shearing stress in the shaft. [5 Marks]
 - (ii) the modulus of rigidity of the material. [4 Marks]
 - (iii) the angle of twist for length AB, BC and CD. [3 Marks]

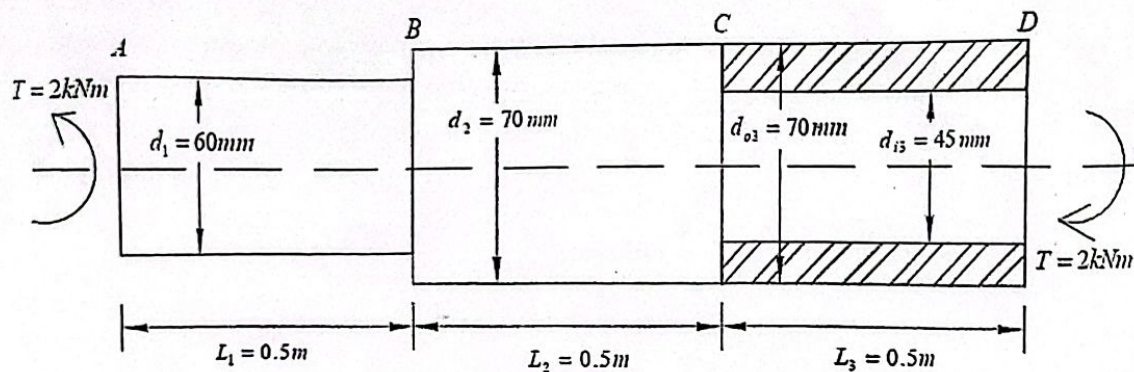


Fig Q3b

QUESTION FOUR (20 Marks)

- (a) A close coiled spring is to have a stiffness of 1kN/m in tension, a maximum load of 50N and a maximum shearing stress of 120MPa. The solid length of the wire (that is, when the coils are touching) is to be 45mm. Find the diameter of the wire, the mean diameter of the coils and the number of the coils required. Take $G = 50\text{GPa}$. You may assume the following equations; [7 Marks]

$$\delta = \frac{8WD^3N}{Gd^4}$$

$$\tau_{\max} = \frac{8WD}{\pi d^3}$$

- (b) An aluminium rod 2.2cm diameter is screwed at the ends and passes through a steel tube of 2.5cm internal diameter and 0.3cm thickness. Both are heated to a temperature of 140°C when the nuts on the rod are screwed tightly on the ends of the tube. Determine the stress on the aluminium rod when the common temperature has fallen to 20°C. For steel $E = 200\text{GPa}$ and $\alpha = 1.2 \times 10^{-5}$ per °C while for aluminium $E = 70\text{GPa}$ and $\alpha = 2.3 \times 10^{-5}$ per °C. [5 Marks]

- (c) Design a helical compression spring to carry a load of 500N with a deflection of 25 mm. The spring index may be taken as 8. Assume the following values for spring material. $\tau = 350\text{N/mm}^2$ and $G = 84\text{kN/mm}^2$. Required: [8 marks]

- (i) wire diameter of spring, d ,
- (ii) mean diameter of the spring coil, D ,
- (iii) Number of active coils
- (iv) Total number of coils
- (v) Solid length L_s
- (vi) Free length L_f
- (vii) Pitch, p

QUESTION FIVE (20 Marks)

- (h) An element of a material in plane two-dimensional stress is subject to direct tensile stresses $\sigma_x = 120 \text{ MN/m}^2$ and $\sigma_y = 80 \text{ MN/m}^2$ that act at 90° to each other and also to shear stress $\tau_{xy} = 50 \text{ MN/m}^2$.
- (i) Determine the stresses that act on this element of the material, when inclined at an angle of 30° . [4 marks]
 - (ii) Show the stresses that act on the faces of the element [2 marks]
- (i) Three strain readings, taken from a *rectangular strain gauge* rosette attached to the surface of a material under stress, were as follows: gauge A, 400μ , gauge B, 300μ and gauge C, 100μ , with gauges B and C being at angles of 45° and 90° measured anticlockwise from gauge A. Given that, for the material, $E = 200 \text{ GN/m}^2$ and Poisson's ratio $\nu = 0.3$, determine the:
- (i) magnitude of the principal strains and the direction of their axes [9 marks]
 - (ii) principal stresses [5 marks]